

TABLE 1 Useful Properties of the Floor and Ceiling Functions

(n is an integer, x is a real number)

(1a) $\lfloor x \rfloor = n$ if and only if $n \leq x < n+1$

(1b) $\lceil x \rceil = n$ if and only if $n-1 < x \leq n$

(1c) $\lfloor x \rfloor = n$ if and only if $x-1 < n \leq x$

(1d) $\lceil x \rceil = n$ if and only if $x \leq n < x+1$

(2) $x-1 < \lfloor x \rfloor \leq x < \lceil x \rceil \leq x+1$

(3a) $\lfloor -x \rfloor = -\lceil x \rceil$

(3b) $\lceil -x \rceil = -\lfloor x \rfloor$

(4a) $\lfloor x+n \rfloor = \lfloor x \rfloor + n$

(4b) $\lceil x+n \rceil = \lceil x \rceil + n$

(1a) $\lfloor x \rfloor = n$ if and only if $n \leq x < n+1$

Proof: Let, $\lfloor x \rfloor = n$. By Defn. 12, n is the largest integer which is less than or equal to x . $\therefore n \leq x$.

We know, $0 < 1$. $\therefore n+0 < n+1 \Rightarrow n < n+1$ $\therefore n \leq x < n+1$

Let, $n \leq x < n+1$. There does not exist any integer in between n and $n+1$ since $(n+1)-n=1$ which is the smallest absolute difference between any two arbitrary different integers. $\therefore$ The largest integer less than or equal to x is n .

$\therefore$ By Defn. 12, $\lfloor x \rfloor = n$. $\square$

(1b) $\lceil x \rceil = n$ if and only if $n-1 < x \leq n$

Proof: Let, $\lceil x \rceil = n$. By Defn. 12, n is the smallest integer that is greater than or equal to x . $\therefore x \leq n$. We know, $-1 < 0 \Rightarrow n-1 < n+0 \Rightarrow n-1 < x \leq n$

$\therefore n-1 < x \leq n$

Let, $n-1 < x \leq n$. There does not exist any integer in between $n-1$ and n , since $n-(n-1)=1$ is the smallest absolute difference between any two arbitrary different integers. $\therefore$ The smallest integer greater than or equal to x is n . By Defn. 12, $\lceil x \rceil = n$. $\square$